The joint code–hull distribution of repeated-root constacyclic codes over finite products of finite fields via transfer matrices

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Abstract

We study λ-constacyclic codes of length $n = Pn_0$, with P a power of the characteristic p and $\gcd(n_0, p) = 1$, over a finite product $A \cong \prod_s K_s$ of finite fields, for twists compatible with the k-Galois pairing. Such codes are parametrized by exponent vectors on the irreducible factors of $x^{n_0} - \mu_s$. We show that the Galois dual and hull are governed by the reciprocal involution on those factors, and that the hull dimension is a sum of nearest-neighbour edge terms $\min(P - \alpha_i, \alpha_{i-1})$ along its orbits. This gives a transfer-matrix formula for the joint distribution of the code dimension and the hull dimension over all labeled exponent assignments; the total number of codes; the number of LCD codes, $2^{\#\mathrm{orbits}}$, independent of P; counts of self-orthogonal and dual-containing codes; and the exact mean and variance of the hull dimension, with orbits of length one treated separately. Worked examples are verified by explicit construction of codes.

Index Terms

Constacyclic codes, repeated-root codes, Galois hull, joint enumerator, transfer matrix, finite fields.

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I. Introduction

For a linear code C over a finite field, the hull — the intersection of C with its dual — controls several constructions, including the amount of entanglement needed in entanglement-assisted quantum codes and the behaviour of LCD (linear complementary dual) codes. Replacing the Euclidean inner product by a Galois inner product [13] leads to the k-Galois hull $\mathrm{Hull}_k(C) = C \cap D(C)$, where $D(C)$ is the dual with respect to a Frobenius-twisted pairing. Galois hulls of constacyclic codes have been determined, at the level of hull dimensions and hull generators, in a number of settings. Over finite fields, Debnath, Prakash and Islam determined the Galois hull dimension of constacyclic codes, including repeated-root lengths $n = n'p^\nu$, and, under the restriction $e \mid 4(e-k)$, $e \nmid 2(e-k)$ on the field degree, the number of such codes with a given Galois hull dimension [9] (see also the correction [10]); average Galois hull dimensions over finite fields and over an auxiliary ring were computed by

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Debnath and Prakash [5], and constacyclic codes with small Galois hull dimension were studied by Debnath, Islam, Yadav and Prakash [8]. For general linear codes, mass formulas counting codes by prescribed hull dimension were obtained in [17] (with the binary case treated in [16]), these counts were shown to decrease as the hull dimension increases in [2], and constructions producing linear codes with prescribed Galois hull dimensions were given in [18]. Over affine algebra rings, Debnath, Islam, Martínez-Moro and Prakash obtained Galois hull formulas for constacyclic codes [7], decomposing the semisimple alphabet into components over $\mathbb{F}_q$ and summing the component hull dimensions, for lengths $n = n'p^{e'}$ under hypotheses on the order of the twist; the enumeration of such codes by hull dimension is left open there. Relative hulls for general linear codes, including e-Galois hulls and their quantum-code consequences, were studied in [1]. Eldin and Solé developed generator polynomial matrices for the Galois hulls of multi-twisted codes, of which constacyclic codes are the index-one case [12]; their results are per-code — a generator polynomial matrix and a hull-dimension formula for a given code, over a single finite field — with no enumeration over codes. Galois hulls of cyclic serial codes over a finite chain ring are studied in [22]; hulls of separable double cyclic codes over $\mathbb{Z}_{p^2}$, including p-dimensions, average p-dimensions and counts at fixed hull p-dimension, were determined in [6], and the existence of cyclic and negacyclic codes with one-dimensional hulls was characterized in [11]; Galois hulls have likewise been used for quantum-code constructions over a finite non-chain ring [24]; repeated-root constacyclic codes and their (Euclidean) duals, including LCD and self-dual enumerations in special lengths, have a substantial literature [4], [21], [23], [25].

a) Contributions and positioning.: In this paper we (1) determine the k-Galois dual and hull of every repeated-root constacyclic code of length Pn_0 over a finite product of finite fields, for compatible twists, in exponent form (Theorems IV.3 and V.2); for a single field component the resulting hull-dimension formula agrees with [9, Theorem 3], and the product-of-fields form given here is what the joint enumerator of item (2) is built on; (2) show that the joint law of $(\dim_{\mathbb{F}_q} C, \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C))$ over labeled assignments is a product of traces of $(P{+}1)$-state transfer matrices indexed by the orbits of the reciprocal involution (Theorem VI.3), with the symmetry $K \leftrightarrow \Delta - K$ (Corollary VI.4); (3) give closed forms for the numbers of LCD, self-orthogonal and dual-containing codes (Corollaries VII.3, VII.4, VII.5 and VII.6); and (4) compute the exact mean and variance of the hull dimension under the uniform labeled model, with separate regimes for orbit lengths 1, 2 and ≥ 3 (Theorems VIII.2 and VIII.4). The transfer-matrix form also has computational content: at numerical values of (u, z), each orbit trace can be evaluated by binary powering, using $O(\log L)$ multiplications of $(P{+}1) \times (P{+}1)$ matrices, whereas direct enumeration inspects all $(P{+}1)^L$ assignments of the orbit. Table I positions these results against the closest earlier work.

The present paper studies a distributional question in the repeated-root setting. Fix a finite product of finite fields $A \cong \prod_{s=1}^{N} K_s$ and a length $n = Pn_0$ in which the characteristic divides n through a prime power P of p, while $\gcd(n_0, p) = 1$. Every constacyclic code over A is then described, component by component, by exponents $a_f \in \{0, \ldots, P\}$ attached to the distinct irreducible factors of a separable polynomial $x^{n_0} - \mu_s$. Rather than the dimension of a single hull, we determine the *joint distribution* of the pair $(\dim_{\mathbb{F}_q} C, \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C))$ over all labeled exponent assignments, for compatible twists. The joint law is captured by a transfer-matrix enumerator: each orbit of the reciprocal involution on the factor set contributes the trace of a power of a $(P{+}1)$-state matrix whose entries record, edge by edge, the code-dimension and hull-dimension contributions (Theorem VI.3). From

TABLE I

POSITIONING RELATIVE TO THE CLOSEST EARLIER WORK. THE COMPARISON WITH [9] IS BASED ON ITS PUBLISHED FULL TEXT (THEOREM 3, COROLLARY 1 AND THEOREM 4); THE CORRECTION [10] MAKES ONLY MINOR TEXTUAL CORRECTIONS AND DOES NOT CHANGE ITS THEOREMS.

Work	Setting	What is determined
[5]	constacyclic codes over finite fields and over an auxiliary ring	average Galois hull dimensions (field case: Theorem 3.3; for one component this is the hull mean of Theorem VIII.2)
[9], [10]	constacyclic codes over a finite field, lengths $n = n'p^{\nu}$	Galois hull dimension of a given code (Theorem 3); number of codes with given hull dimension when reciprocal-orbit lengths are $1, 2$ and 4 (Theorem 4)
[7]	constacyclic codes over affine algebra rings (components over $\mathbb{F}_q$), lengths $n = n'p^{e'}$	Galois hull dimension of a given code, summed over components; LCD criterion; entanglement-assisted constructions; enumeration by hull dimension left open
[12]	multi-twisted codes over a finite field, of which constacyclic codes are the index-one case	generator polynomial matrix and dimension formula for the Galois hull of a given code; self-orthogonal and LCD criteria; no enumeration over codes
This paper	repeated-root length $n = Pn_0$ with compatible twists, over a finite product of finite fields	joint distribution of $(\dim C, \dim \mathrm{Hull}_k(C))$; LCD, self-orthogonal and dual-containing counts; exact mean and variance

the enumerator we extract the total number of codes, the code-dimension distribution, the number of LCD codes, characterizations and counts of self-orthogonal and dual-containing codes, and exact first and second moments of the hull dimension. Orbits of length one enter through a separate constant: their hull edge is a self-loop $\min(P-A, A)$, with mean $\nu_P = \lfloor P^2/4 \rfloor/(P+1)$, rather than the ordinary-edge mean $\mu_P = P(2P+1)/(6(P+1))$; the two constants must not be interchanged when a fixed factor of the reciprocal involution is present.

The paper is organized as follows. Section II fixes the conventions, in particular the code-first Galois pairing on which the dual formulas depend. Sections III–V develop the parametrization, the Galois dual, and the hull formula in exponent form. Section VI proves the transfer-matrix enumerator and its specialization at $P = 1$. Sections VII and VIII contain the counting corollaries and the moment formulas. Section IX gives worked examples and describes the computational verification; Section X records the scope and limitations. Throughout, counts are over labeled exponent assignments, not over equivalence or isometry classes of codes.

II. PRELIMINARIES AND CONVENTIONS

Let p be a prime, $q = p^e$, and let $0 \le k < e$ be a fixed Galois index. Let

$$A \;\cong\; \prod_{s=1}^{N} K_s, \qquad K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}, \quad d_s = em_s,$$

with fixed component labels. (Standard terminology for constacyclic codes follows [14], [19].)

Terminology. We call A an *affine algebra*: equivalently, A is a finite-dimensional commutative $\mathbb{F}_q$-algebra of the form $\mathbb{F}_q[X_1, \dots, X_\ell]/\langle t_1(X_1), \dots, t_\ell(X_\ell)\rangle$ in which each t_i is *square-free*, i.e. has no repeated irreducible factor;

square-freeness is what makes A reduced (étale), hence a finite product of finite fields. Only this decomposition, the component Galois twists, and the weights $w_O = m_s \deg f$ enter the results below, so we state the setting throughout as a finite product of finite fields. A *labeled* assignment is an exponent tuple together with the fixed component and factor labels; all counts are over labeled assignments. On K_s write $\sigma(\alpha) = \alpha^{p^k}$ and $\rho(\alpha) = \alpha^{p^{d_s-k}} = \sigma^{-1}$.

Definition II.1 (code-first Galois pairing)**.** For one component (the subscript s suppressed), the k-Galois pairing on K^n is $\langle x, y\rangle_k = \sum_{i=0}^{n-1} x_i\, \sigma(y_i)$, the *code-first* dual of a code $C \subseteq K^n$ is $D(C) = \{y \in K^n : \langle c, y\rangle_k = 0 \text{ for all } c \in C\}$, and the k-Galois hull is $\mathrm{Hull}_k(C) = C \cap D(C)$. The Euclidean dual is denoted $C^\perp$.

This is the k-Galois inner product introduced by Fan and Zhang [13]. The code-first slot order agrees with the Galois dual convention of Liu and Pan [15]; in the finite-field case the dictionary is $D(C) = \sigma^{e-k}(C^\perp)$. The slot order is part of the definition: the candidate y occupies the σ-twisted slot, and this is what makes $D(C) = \rho(C^\perp)$ (coordinatewise ρ; Proposition IV.2) rather than the analogous statement with σ. All dual formulas below use this convention. The global pairing on A^n is the tuple of component pairings; a code over A is an A-submodule of A^n.

Fix a power P of p, and $n = Pn_0$ with $\gcd(n_0, p) = 1$. Let $\lambda = (\lambda_s) \in A^*$ and $M_s(x) = x^n - \lambda_s \in K_s[x]$. Since $a \mapsto a^P$ is a bijection of the finite field K_s, there is a unique $\mu_s \in K_s^*$ with $\mu_s^P = \lambda_s$. Writing $\bar{M}_s(x) = x^{n_0} - \mu_s$, we have $M_s = \bar{M}_s^P$ in characteristic p, and $\bar{M}_s$ is separable because $\bar{M}_s' = n_0 x^{n_0-1}$ and every root of $\bar{M}_s$ is nonzero. Let $\mathcal{F}_s$ be the set of distinct monic irreducible factors of $\bar{M}_s$ over K_s; then $M_s = \prod_{f\in\mathcal{F}_s} f^P$.

An *exponent assignment* on component s is a tuple $a = (a_f)_{f\in\mathcal{F}_s}$ with $a_f \in \{0, \dots, P\}$; put $g_a = \prod_f f^{a_f}$ and $C_s(a) = \langle g_a\rangle \subseteq K_s[x]/(M_s)$. A global assignment is a tuple of component assignments.

A λ-*constacyclic code* over A is an A-submodule of A^n closed under the componentwise λ-constacyclic shift $(c_0, \dots, c_{n-1}) \mapsto (\lambda c_{n-1}, c_0, \dots, c_{n-2})$. Under the coefficient identification $A^n \cong A[x]/(x^n - \lambda)$, this shift is multiplication by x; hence the λ-constacyclic codes over A are exactly the ideals of $A[x]/(x^n - \lambda)$, an equivalence that needs no separability or compatibility hypothesis. From now on, "code over A" means a λ-constacyclic code over A; componentwise these are tuples of λ_s-constacyclic codes (Proposition VI.2).

For a monic polynomial $f(x) = \sum_{i=0}^{r} c_i x^i$ with $f(0) = c_0 \neq 0$, define the (Frobenius-twisted) reciprocal

$$f^{\#}(x) \;=\; \rho(c_0)^{-1} \sum_{i=0}^{r} \rho(c_i)\, x^{\,r-i},$$

which is monic of degree r. Every $f \in \mathcal{F}_s$ satisfies $f(0) \neq 0$ since $\bar{M}_s(0) = -\mu_s \neq 0$.

Definition II.2 (compatibility and the involution τ_s)**.** The twist λ_s is *compatible* if $\lambda_s^{1+p^{d_s-k}} = 1$. Since K_s^* is cyclic, the number of compatible twists in K_s^* is $\gcd\left(1 + p^{d_s-k},\, p^{d_s} - 1\right)$; in particular $\lambda_s = 1$ is always compatible, and $\lambda_s = -1$ is compatible when p is odd. Under compatibility, $\tau_s : \mathcal{F}_s \to \mathcal{F}_s$, $\tau_s(f) = f^{\#}$, is a permutation (Theorem III.2). We assume compatibility for every s from Section IV on. A τ_s-orbit is written $O = (f_0, \dots, f_{L-1})$ with $\tau_s(f_i) = f_{i+1}$, indices modulo $L = L_O$; we set $\alpha_i = a_{f_i}$ and call α_{i-1} the predecessor of α_i. Since $\#$ preserves degrees, all factors in an orbit have the same degree; the *weight* of O and of a factor f is $w_O = m_s \deg_{K_s} f$ and $w_f = m_s \deg_{K_s} f$.

Dimensions over $\mathbb{F}_q$ are recovered from dimensions over K_s by $\dim_{\mathbb{F}_q} V = m_s \dim_{K_s} V$. We write u, z for indeterminates and work in $\mathbb{Z}[u, z]$.

III. Parametrization and the reciprocal operation

Theorem III.1 (parametrization)**.** *The map $a \mapsto C_s(a) = \langle g_a \rangle$ is a bijection from $\{0, \dots, P\}^{\mathcal{F}_s}$ onto the set of λ_s-constacyclic codes of length n over K_s, and $\dim_{K_s} C_s(a) = n - \sum_f a_f \deg f$. Consequently there are $(P+1)^{\sum_s |\mathcal{F}_s|}$ labeled global assignments, and λ-constacyclic codes over A correspond to tuples of component assignments (Proposition VI.2).*

Proof. Ideals of $K_s[x]/(M_s)$ correspond to ideals $(M_s) \subseteq I$ of the principal ideal domain $K_s[x]$, hence to monic divisors $\bar{g}$ of M_s; by unique factorization and $M_s = \prod f^P$, monic divisors are exactly the g_a. For a monic divisor g of degree r, multiplication by g identifies $\langle g \rangle \subseteq K_s[x]/(M_s)$ with $K_s[x]/((M_s)/g)$, of dimension $n - r$. No separability hypothesis is used. □

Theorem III.2 (the reciprocal operation)**.**

(i) *$\#$ is multiplicative on monic polynomials with nonzero constant term; hence $\left(\prod_f f^{a_f}\right)^{\#} = \prod_f (f^{\#})^{a_f}$ for all exponents $a_f \geq 0$.*

(ii) *If $f(\xi) = 0$ with $\xi \neq 0$ in an algebraic closure, then $f^{\#}\left(\xi^{-p^{d_s - k}}\right) = 0$; in fact $f^{\#}$ is the minimal polynomial of $\xi^{-p^{d_s-k}}$.*

(iii) *$\lambda_s^{1+p^{d_s-k}} = 1 \iff \mu_s^{1+p^{d_s-k}} = 1$. Under compatibility, τ_s permutes $\mathcal{F}_s$ and preserves degrees.*

Proof. Let $\bar{\rho}$ be the coefficientwise extension of ρ to $K_s[x]$; it is a ring automorphism with inverse $\bar{\sigma}$. For (i), $\overline{\rho}(fg) = \bar{\rho}(f)\bar{\rho}(g)$, reversal $x^r f(1/x)$ is multiplicative in fixed degrees, and the normalizing constants multiply since $(fg)_0 = f_0 g_0$; combining the three gives $(fg)^{\#} = f^{\#} g^{\#}$, and induction on the number of factors gives the displayed identity. For (ii), apply the Frobenius map $t \mapsto t^{p^{d_s-k}}$ of the algebraic closure to $\sum c_i \xi^i = 0$ to get $\sum \rho(c_i) \beta^{-i} = 0$ for $\beta = \xi^{-p^{d_s-k}}$; multiplying by β^r yields $\rho(c_0) f^{\#}(\beta) = 0$. The same Frobenius transport preserves degrees of minimal polynomials, so $f^{\#}$ is the (monic irreducible) minimal polynomial of β. For (iii), $(\mu_s^{1+p^{d_s-k}})^P = \lambda_s^{1+p^{d_s-k}}$ and in characteristic p, $y^P = 1$ implies $y = 1$; now (ii) shows $\tau_s(f)$ is the minimal polynomial of a root β with $\beta^{n_0} = \rho(\mu_s)^{-1} = \mu_s$ (using compatibility in the form $\rho(\mu_s) = \mu_s^{-1}$), so $\tau_s(f) \mid \bar{M}_s$; bijectivity follows by applying the same construction with σ in place of ρ, which gives the inverse map. □

IV. The Galois dual

The first step is a Euclidean-dual statement that uses no separability at all.

Lemma IV.1 (Euclidean dual generator)**.** *Let g be a monic divisor of $x^n - \lambda$, $\lambda \in K^*$, $h = (x^n - \lambda)/g$, $r = \deg g$, $s = n - r$, and $C = \langle g \rangle$ under the coefficient identification (this is the classical constacyclic dual statement; cf. [14], [19]). Then $h_0 \neq 0$, and with $h^{\natural}(x) = h_0^{-1} \sum_{i=0}^{s} h_i x^{s-i}$ (the ordinary reciprocal, without Frobenius), $C^{\perp} = \langle h^{\natural} \rangle$, a λ^{-1}-constacyclic code.*

Proof. From $gh = x^n - \lambda$, $g_0 h_0 = -\lambda \neq 0$. Both $\langle h^{\natural} \rangle$ and $C^{\perp}$ have dimension r: the first by the divisor dimension count of Theorem III.1 (applied to any modulus), the second by rank–nullity. The boundary cases are

immediate from the definitions: if $r = 0$, then $g = 1$, $C = K^n$ and $C^\perp = \{0\}$, while $h^\natural$ has degree n and hence no nonzero multiple of degree $< n$; if $s = 0$, then $h = 1$, $C = \{0\}$ and $C^\perp = K^n$, while $h^\natural = 1$. Assume $r, s \geq 1$ below, so that V is defined (and $v = 0$ gives $d = 0$). We show orthogonality. For polynomials a, d of degree $< n$, write $\mathrm{rev}_n(d) = x^{n-1}d(1/x)$; then $\sum_i a_i d_i = [x^{n-1}]\big(a(x)\,\mathrm{rev}_n(d)(x)\big)$. Take $a = gu$, $\deg u < s$, and $d = h^\natural v$, $\deg v < r$. Since $h^\natural(1/x) = h_0^{-1}x^{-s}h(x)$, one computes $\mathrm{rev}_n(d) = h_0^{-1}h(x)V(x)$ with $V(x) = x^{r-1}v(1/x)$, $\deg V \leq r-1$. Hence $a(x)\,\mathrm{rev}_n(d)(x) = h_0^{-1}(x^n - \lambda)u(x)V(x)$ with $\deg(uV) \leq n-2$; the coefficient of x^{n-1} is 0. Thus $\langle h^\natural\rangle \subseteq C^\perp$, and equality follows by dimensions. That $h^\natural \mid x^n - \lambda^{-1}$ follows from $(gh)^* = g^*h^* = -\lambda(x^n - \lambda^{-1})$ for the unnormalized reciprocal $\phi^*(x) = x^{\deg\phi}\phi(1/x)$. □

Proposition IV.2 (transport by ρ)**.** *Let the field be $\mathbb{F}_{p^d}$, with σ, ρ as in Section II. For every $\mathbb{F}_{p^d}$-linear code $C \subseteq \mathbb{F}_{p^d}^n$, $D(C) = \rho(C^\perp)$ (coordinatewise). Consequently, for $C = \langle g\rangle \subseteq \mathbb{F}_{p^d}[x]/(x^n - \lambda)$ and $h = (x^n - \lambda)/g$, $D(C) = \langle h^\#\rangle \subseteq \mathbb{F}_{p^d}[x]/(x^n - \lambda^{-p^{d-k}})$; if λ is compatible, the modulus is again $x^n - \lambda$.*

Proof. By the code-first definition, $y \in D(C)$ iff $\sum_i c_i\sigma(y_i) = 0$ for all $c \in C$, i.e. iff $\sigma(y) \in C^\perp$ (coordinatewise), i.e. iff $y \in \rho(C^\perp)$ since $\rho = \sigma^{-1}$; this is exactly where the slot order enters. Coefficientwise application of an automorphism sends the ideal $\langle e\rangle$ in $K[x]/(x^n - \eta)$ to $\langle\bar\rho(e)\rangle$ in $K[x]/(x^n - \rho(\eta))$; applying this to Lemma IV.1 and comparing $\bar\rho(h^\natural)$ with $h^\#$ gives the generator. Compatibility $\lambda^{1+p^{d-k}} = 1$ gives $\lambda^{-p^{d-k}} = \lambda$. □

Theorem IV.3 (dual exponents)**.** *For every assignment a, with $h_a = M_s/g_a$, $D(C_s(a)) = \langle h_a^\#\rangle$; under compatibility this is a λ_s-constacyclic code in the same quotient. On a τ_s-orbit $O = (f_0, \ldots, f_{L-1})$ with $\alpha_i = a_{f_i}$, the exponent of f_i in $h_a^\#$ is*

$$\beta_i \;=\; P - \alpha_{i-1} \qquad (\textit{indices mod } L).$$

Proof. The generator statement is Proposition IV.2. For the exponents, $h_a = \prod_f f^{P-a_f}$, so by Theorem III.2(i) and the permutation property, $h_a^\# = \prod_f(\tau_s f)^{P-a_f} = \prod_g g^{P-a_{\tau_s^{-1}g}}$; for $g = f_i$, $\tau_s^{-1}(f_i) = f_{i-1}$, giving $P - \alpha_{i-1}$. □

Remark. Without compatibility, $D(C_s(a)) = \langle h_a^\#\rangle$ and the exponent formula remain valid, but in the modulus $x^n - \lambda_s^{-p^{d_s-k}}$. Compatibility is what allows the dual to be viewed inside the same constacyclic family; it cannot simply be dropped (e.g. in $\mathbb{F}_9$ with $k = 1$, a twist λ of order 8 satisfies $\lambda^{1+3} = \lambda^4 = -1 \neq 1$ and indeed $\lambda^{-3} \neq \lambda$).

V. The Galois hull

Lemma V.1 (intersection of ideals)**.** *For monic divisors g, l of a monic $M \in K[x]$, the images of (g) and (l) in $K[x]/(M)$ intersect in the image of $(\mathrm{lcm}(g,l))$, and $\mathrm{lcm}(g,l) = \prod_\pi \pi^{\max(v_\pi(g), v_\pi(l))}$.*

Proof. In $K[x]$, $(g) \cap (l) = (\mathrm{lcm}(g,l))$ since both consist of the common multiples of g and l, characterized valuation-wise by the displayed maximum. If y lies in both images in the quotient, any lift Y satisfies $Y \in (g)$ and $Y \in (l)$ because $(M) \subseteq (g), (l)$; hence $Y \in (\mathrm{lcm}(g,l))$. The reverse inclusion is immediate from $g, l \mid \mathrm{lcm}(g,l)$. □

Theorem V.2 (hull generator and dimensions)**.** *Under compatibility, for every assignment a,* $\mathrm{Hull}_k(C_s(a)) = \langle \mathrm{lcm}(g_a, h_a^{\#}) \rangle$*, the exponent of f_i in the hull generator being* $\max(\alpha_i, P - \alpha_{i-1})$*. Moreover*

$$\dim_{\mathbb{F}_q} C(a) \;=\; \sum_s \sum_O w_O \sum_i (P - \alpha_i), \qquad \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C(a)) \;=\; \sum_s \sum_O w_O \sum_i \min(P - \alpha_i,\, \alpha_{i-1}),$$

the inner sums over orbit positions (indices mod L_O); here and below, α_i refers to the orbit (s, O) under discussion.

Proof. Work in one component. By Theorems III.1 and IV.3, $C(a)$ and $D(C(a))$ are the images of (g_a) and $(h_a^{\#})$, both dividing $M = x^n - \lambda$; Lemma V.1 gives the hull generator, whose f_i-exponent is $\max(\alpha_i, \beta_i) = \max(\alpha_i, P - \alpha_{i-1})$. Dimensions are $n - \deg(\text{generator})$ over K; grouping factors by orbits (constant degree on an orbit) and using $P - \max(u, v) = \min(P - u, P - v)$ with $v = P - \alpha_{i-1}$ gives the hull formula; multiplication by m_s and summation over s gives the $\mathbb{F}_q$-dimensions. □

Remark (relation to [9]). For a single component ($N = 1$), Theorem V.2 recovers [9, Theorem 3]: with $P = p^\nu$ and cycle contribution $a_t P - \sum_i \max(u_i, P - u_{i-1})$, the identity $P - \max(u, P - v) = \min(P - u, v)$ identifies their summand with $\sum_i \min(P - \alpha_i, \alpha_{i-1})$ above, the factor degree $\mathrm{ord}_j(q)$ being our orbit weight w_O. Their counting result [9, Theorem 4] is restricted to reciprocal-orbit lengths $1, 2$ and 4 and counts by hull dimension alone; the joint enumerator below imposes no restriction on orbit length and records both dimensions.

VI. The transfer-matrix enumerator

For $P \geq 1$, $w \geq 1$, define the $(P{+}1) \times (P{+}1)$ matrix over $\mathbb{Z}[u, z]$

$$T_w^{(P)}(u, z)_{a,b} \;=\; u^{\,w(P-b)}\, z^{\,w \min(P-b,\, a)}, \qquad a, b \in \{0, \ldots, P\},$$

the row index a being the previous exponent and the column index b the current one.

Theorem VI.1 (local transfer trace)**.** *For a τ_s-orbit O of length L and weight w,*

$$\mathrm{tr}\big(T_w^{(P)}(u, z)^L\big) \;=\; \sum_{\alpha \in \{0,\ldots,P\}^L} u^{\,w \sum_i (P - \alpha_i)}\, z^{\,w \sum_i \min(P - \alpha_i,\, \alpha_{i-1})}, \quad \textit{indices mod } L.$$

Proof. Expanding the matrix power, $(T^L)_{a_0, a_L} = \sum_{a_1, \ldots, a_{L-1}} \prod_{j=1}^{L} T_{a_{j-1}, a_j}$; setting $a_L = a_0$ and summing over a_0 gives $\mathrm{tr}(T^L) = \sum_{a_0, \ldots, a_{L-1}} \prod_{j=1}^{L} T_{a_{j-1}, a_j}$. These closed walks are in bijection with assignments $\alpha = (\alpha_0, \ldots, \alpha_{L-1})$ (orbit positions are labeled; a walk and its rotation are different assignments). The j-th edge contributes $u^{w(P - a_j)} z^{w \min(P - a_j, a_{j-1})}$; summing exponents over the walk and reindexing gives the claim. For $L = 1$ the single edge is the self-loop $a_0 \to a_0$ and the formula reads $\sum_a u^{w(P-a)} z^{w \min(P-a, a)}$, as required. □

Proposition VI.2 (product assembly)**.** *Ideals of $A[x]/(x^n - \lambda) \cong \prod_s K_s[x]/(x^n - \lambda_s)$ are exactly the products of component ideals; the global dual and hull decompose componentwise; and* $\dim_{\mathbb{F}_q} C = \sum_s m_s \dim_{K_s} C_s$*, likewise for the hull.*

Proof. The ring isomorphism is coefficientwise; quotienting a product by the product ideal $(x^n - \lambda_s)_s$ gives the product of quotients (this is the direct-product structure of the square-free algebra A itself; no coprimality between components is involved). Ideals of a finite product of rings are products of ideals (project, and recover the components

with the idempotents of A). The pairing is componentwise and vanishes exactly when every component does, giving the dual and hull statements; dimensions add over the direct sum $A^n \cong \bigoplus_s K_s^n$ with the factor $[K_s : \mathbb{F}_q] = m_s$. □

Theorem VI.3 (joint enumerator)**.** *Under the standing hypotheses (common P, n, k; compatible twists; labeled components),*

$$\mathcal{E}(u,z) \;:=\; \sum_{\text{labeled } a} u^{\dim_{\mathbb{F}_q} C(a)} z^{\dim_{\mathbb{F}_q} \mathrm{Hull}_k(C(a))} \;=\; \prod_{s=1}^{N} \prod_{O \in \mathcal{O}_s} \mathrm{tr}\Big(T_{w_O}^{(P)}(u,z)^{L_O}\Big),$$

where $\mathcal{O}_s$ is the set of τ_s-orbits. The coefficient of $u^K z^H$ is the number of labeled assignments with $\dim_{\mathbb{F}_q} C = K$ and $\dim_{\mathbb{F}_q} \mathrm{Hull}_k(C) = H$.

Proof. By Theorem III.1 and Proposition VI.2, a global assignment is the disjoint union of its orbit restrictions $\alpha^{(O)} \in \{0, \dots, P\}^{L_O}$, and by Theorem V.2 each of the two dimensions is a sum over (s, O) of a quantity depending only on $\alpha^{(O)}$. The finite distributive law (expanding a product of sums over a Cartesian product index set) factorizes the global sum into the product of the orbit sums $\mathcal{E}_O(u,z) = \sum_\alpha u^{w_O \sum (P-\alpha_i)} z^{w_O \sum \min(P-\alpha_i, \alpha_{i-1})}$, and Theorem VI.1 identifies $\mathcal{E}_O(u,z) = \mathrm{tr}(T_{w_O}^{(P)}(u,z)^{L_O})$. □

Corollary VI.4 (joint symmetry of the enumerator)**.** *Put $\Delta := \sum_{s,O} P w_O L_O = n \sum_s m_s$ (indeed, for each component s, $\sum_{O \in \mathcal{O}_s} w_O L_O = m_s n_0$, since the degrees of the distinct irreducible factors of $\bar{M}_s$ sum to n_0; hence Δ is the maximum, and ambient, $\mathbb{F}_q$-dimension). Then for all integers K, H,*

$$\#\{a : (\dim_{\mathbb{F}_q} C(a), \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C(a))) = (K,H)\} = \#\{a : (\dim_{\mathbb{F}_q} C(a), \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C(a))) = (\Delta - K, H)\},$$

equivalently $\mathcal{E}(u,z) = u^\Delta \mathcal{E}(u^{-1}, z)$. In particular, for every K the multiset of hull dimensions over the assignments with $\dim_{\mathbb{F}_q} C = K$ equals that over the assignments with $\dim_{\mathbb{F}_q} C = \Delta - K$.

Proof. It suffices to argue on one orbit O of length L and weight w; the global statement follows by summing over orbits. Define $\alpha'_i = P - \alpha_{-i}$ (indices mod L). This is an involution of $\{0, \dots, P\}^L$: applying it twice returns α_i. Its code contribution is $w \sum_i (P - \alpha'_i) = w \sum_i \alpha_{-i} = w \sum_i \alpha_i = PwL - w \sum_i (P - \alpha_i)$, so K maps to $PwL - K$. Its hull contribution satisfies, with the reindexing $j = -i + 1$, $\min(P - \alpha'_i, \alpha'_{i-1}) = \min(\alpha_{-i}, P - \alpha_{-i+1}) = \min(P - \alpha_j, \alpha_{j-1})$, term by term equal to the hull sum of α. Hence the involution pairs tuples of joint type (K, H) with tuples of type $(PwL - K, H)$, proving the per-orbit coefficient identity; the product over all orbits (Theorem VI.3) gives the claim with $\Delta = \sum_O P w_O L_O$. □

Remark (hypotheses of the identity). As an identity in $\mathbb{Z}[u, z]$, Theorem VI.1 holds for every integer $P \geq 1$; that P is a power of p is needed for the field construction behind Theorems III.1–V.2, not for the matrix identity. The enumerator counts labeled assignments; the trace formula is insensitive to reversing the orbit orientation. Indeed the hull dimension of every individual assignment is unchanged under reversal, because $\min(P-a, b) - \min(P-b, a) = b - a$ telescopes to 0 around the orbit; hence no statistic of Sections VII and VIII can detect the orientation. The predecessor convention α_{i-1} in Theorem V.2 is nevertheless essential for identifying the hull *generator*, and is fixed by the code-first pairing of Section II.

Proposition VI.5 (simple-root recovery at $P = 1$)**.** *At $P = 1$, assignments are binary, g_a is a squarefree divisor, the hull edge term $\min(1 - \alpha_i, \alpha_{i-1})$ is the indicator of the transition $1 \to 0$ along the orbit, and*

$$T_w^{(1)}(u, z) \;=\; \begin{pmatrix} u^w & 1 \\ u^w z^w & 1 \end{pmatrix} \qquad (\textit{rows/columns } a, b \in \{0, 1\}),$$

so Theorem VI.3 specializes to the binary-state simple-root joint enumerator with the same conventions.

Proof. For $P = 1$, $\min(1 - b, a)$ is 1 exactly for $(a, b) = (1, 0)$; reading off the four entries gives the matrix, and $\mu_s = \lambda_s$, $\bar{M}_s = M_s$ gives the squarefree parametrization. □

VII. Counting corollaries

Corollary VII.1 (total count)**.** $\mathcal{E}(1, 1) = (P + 1)^{\sum_s |\mathcal{F}_s|}$.

Proof. At $u = z = 1$ each $T_w^{(P)}$ is the all-ones matrix J with $J^L = (P + 1)^{L-1} J$, so $\operatorname{tr}(J^{L_O}) = (P + 1)^{L_O}$; product over orbits gives $(P + 1)^{\sum_s \sum_O L_O} = (P + 1)^{\sum_s |\mathcal{F}_s|}$. □

Corollary VII.2 (code-dimension distribution)**.** $\mathcal{E}(u, 1) = \displaystyle\prod_s \prod_{f \in \mathcal{F}_s} \big(1 + u^{w_f} + \cdots + u^{P w_f}\big)$.

Proof. At $z = 1$ the entry $T_{a,b} = u^{w(P-b)}$ is independent of the row, so $T = c\, r$ is rank one with $r \cdot c = \sum_{j=0}^{P} u^{wj}$; then $\operatorname{tr}(T^L) = (r \cdot c)^{L-1} \operatorname{tr}(T) = \big(\sum_{j=0}^{P} u^{wj}\big)^L$. Within an orbit all w_f equal w_O, giving $\prod_{f \in O}(1 + u^{w_f} + \cdots + u^{P w_f})$ per orbit. □

Corollary VII.3 (LCD count)**.** *The number of labeled assignments with $\operatorname{Hull}_k(C(a)) = 0$ is $2^{\#\textit{orbits}}$, independent of P.*

Proof. The hull dimension is a sum of nonnegative terms $w_O \min(P - \alpha_i, \alpha_{i-1})$; it vanishes iff every orbit satisfies $\min(P - \alpha_i, \alpha_{i-1}) = 0$ for all i. On one orbit, if some $\alpha_m \notin \{0, P\}$ (so $P \geq 2$), then $\min(P - \alpha_{m+1}, \alpha_m) = 0$ with $\alpha_m > 0$ forces $\alpha_{m+1} = P$, and inductively every $\alpha_i = P$ around the cycle, a contradiction (for $L = 1$, $\min(P - \alpha_0, \alpha_0) = 0$ already forces $\alpha_0 \in \{0, P\}$). Hence each $\alpha_i \in \{0, P\}$; if both values occurred, at a boundary position with $\alpha_i = 0$, $\alpha_{i-1} = P$ the term would be $\min(P, P) = P \geq 1$. So exactly two tuples per orbit ($\alpha \equiv 0$ or $\alpha \equiv P$), independently across orbits. □

Corollary VII.4 (LCD refinement: the code dimension of LCD assignments)**.** *With Δ as in Corollary VI.4,*

$$\sum_{a \,:\, \operatorname{Hull}_k(C(a)) = 0} u^{\dim_{\mathbb{F}_q} C(a)} \;=\; \prod_s \prod_{O \in \mathcal{O}_s} \big(1 + u^{P w_O L_O}\big).$$

Consequently, under the uniform distribution on labeled assignments, conditionally on $\operatorname{Hull}_k(C(a)) = 0$ the code dimension is distributed as $\sum_O (P w_O L_O)\, \xi_O$ with independent $\xi_O \sim \operatorname{Bernoulli}(1/2)$; in particular $\mathbb{E}[\dim_{\mathbb{F}_q} C \mid \operatorname{Hull}_k(C) = 0] = \Delta/2$ and $\operatorname{Var}(\dim_{\mathbb{F}_q} C \mid \operatorname{Hull}_k(C) = 0) = \frac{P^2}{4} \sum_O (w_O L_O)^2$.

Proof. By Corollary VII.3, $\operatorname{Hull}_k(C(a)) = 0$ if and only if every orbit restriction is constant, $\alpha \equiv 0$ or $\alpha \equiv P$; these two tuples have code contributions $P w_O L_O$ and 0 respectively (Theorem V.2), giving the displayed product by

independence across orbits. Conditioning the uniform distribution on this product event keeps the orbits independent with the two options equally likely, which is the stated Bernoulli sum; its mean is $\frac{1}{2}\sum_O Pw_OL_O = \Delta/2$ and its variance is $\sum_O (Pw_OL_O)^2 \cdot \frac{1}{4}$. □

Corollary VII.5 (self-orthogonal and dual-containing codes)**.** *$C(a) \subseteq D(C(a))$ if and only if $\alpha_{i-1} + \alpha_i \geq P$ on every orbit edge, and then the count is $\prod_O \operatorname{tr}(B_P^{L_O})$ with $(B_P)_{a,b} = 1$ iff $a + b \geq P$. Dually, $D(C(a)) \subseteq C(a)$ if and only if $\alpha_{i-1} + \alpha_i \leq P$ on every edge, with count $\prod_O \operatorname{tr}((B'_P)^{L_O})$, $(B'_P)_{a,b} = 1$ iff $a + b \leq P$. The two counts are equal.*

Proof. Since $\mathrm{Hull} = C \cap D(C) \subseteq C$, $C \subseteq D(C)$ iff $\dim \mathrm{Hull} = \dim C$. By Theorem V.2, $\dim C - \dim \mathrm{Hull} = \sum_{s,O} w_O \sum_i \max(0, P - \alpha_i - \alpha_{i-1})$ because $(P - \alpha_i) - \min(P - \alpha_i, \alpha_{i-1}) = \max(0, P - \alpha_i - \alpha_{i-1})$; every summand is nonnegative, so the total vanishes iff every edge satisfies $\alpha_{i-1} + \alpha_i \geq P$. The admissible tuples on one orbit are the closed walks of the 0-1 matrix B_P, counted by $\operatorname{tr}(B_P^{L_O})$ as in Theorem VI.1. For the dual side, $\dim D(C) = \sum w_O \sum_i \alpha_{i-1} = \sum w_O \sum_i \alpha_i$ by Theorem IV.3, and $\alpha_{i-1} - \min(P - \alpha_i, \alpha_{i-1}) = \max(0, \alpha_i + \alpha_{i-1} - P)$; the same argument gives the condition $\alpha_{i-1} + \alpha_i \leq P$ and the count $\operatorname{tr}((B'_P)^{L_O})$. The involution $\alpha_i \mapsto P - \alpha_i$ maps one edge condition to the other, so the counts agree. □

Corollary VII.6 (restricted enumerators for the self-orthogonal and dual-containing families)**.** *For an orbit of length L and weight w, define matrices over $\mathbb{Z}[t]$ and $\mathbb{Z}[u, z]$ by*

$$\big(A_w^{\mathrm{SO}}(t)\big)_{a,b} = \begin{cases} t^{\,w(P-b)}, & a + b \geq P, \\ 0, & a + b < P, \end{cases} \qquad \big(A_w^{\mathrm{DC}}(u,z)\big)_{a,b} = \begin{cases} u^{\,w(P-b)} z^{\,wa}, & a + b \leq P, \\ 0, & a + b > P, \end{cases}$$

the row index a being the previous exponent and the column index b the current one, as in Theorem VI.1. Then

$$\sum_{a\,:\,C(a) \subseteq D(C(a))} t^{\dim_{\mathbb{F}_q} C(a)} = \prod_s \prod_{O \in \mathcal{O}_s} \operatorname{tr}\Big(\big(A_{w_O}^{\mathrm{SO}}(t)\big)^{L_O}\Big),$$

whose specialization $t = uz$ is the joint enumerator of the self-orthogonal family (there $\dim_{\mathbb{F}_q} \mathrm{Hull}_k = \dim_{\mathbb{F}_q} C$), and

$$\sum_{a\,:\,D(C(a)) \subseteq C(a)} u^{\dim_{\mathbb{F}_q} C(a)} z^{\dim_{\mathbb{F}_q} \mathrm{Hull}_k(C(a))} = \prod_s \prod_{O \in \mathcal{O}_s} \operatorname{tr}\Big(\big(A_{w_O}^{\mathrm{DC}}(u,z)\big)^{L_O}\Big).$$

Writing $N_{\mathrm{SO}}(K)$ for the number of self-orthogonal assignments with $\dim_{\mathbb{F}_q} C = K$ and $N_{\mathrm{DC}}(K, H)$ for the number of dual-containing assignments with $(\dim_{\mathbb{F}_q} C, \dim_{\mathbb{F}_q} \mathrm{Hull}_k) = (K, H)$, one has $N_{\mathrm{SO}}(K) = N_{\mathrm{DC}}(\Delta - K, K)$ for all K. At $u = z = 1$ the two enumerators reduce to the two counts of Corollary VII.5.

Proof. Both admissibility conditions are those of Corollary VII.5, and both dimensions are sums over orbits, so by the closed-walk expansion of Theorem VI.1 it suffices to check one orbit. On a self-orthogonal tuple, $\mathrm{Hull}_k(C) = C$, so the joint weight is $(uz)^{\dim_{\mathbb{F}_q} C}$; along an edge $a \to b$ the code contribution is $w(P - b)$ and admissibility is $a + b \geq P$, which is exactly the weight built into A_w^{SO}. On a dual-containing tuple, $\mathrm{Hull}_k(C) = D(C)$, whose per-orbit dimension is $w \sum_i \alpha_i = w \sum_i \alpha_{i-1}$ (proof of Corollary VII.5); the edge $a \to b$ therefore carries code weight $w(P - b)$ (from b) and hull weight wa (from the previous exponent), with admissibility $a + b \leq P$, as in A_w^{DC}. The trace includes the closing edge, i.e. the orbit-boundary condition. For the symmetry, the per-orbit

complement $\alpha_i \mapsto P - \alpha_i$ maps the self-orthogonal condition to the dual-containing condition bijectively, sends $\dim_{\mathbb{F}_q} C = K$ to $\dim_{\mathbb{F}_q} C = \Delta - K$, and sends the hull dimension to $\sum_O w_O \sum_i (P - \alpha_i) = K$. □

VIII. Moments of the hull dimension

The model. By Theorem III.1, labeled assignments are in bijection with the labeled constacyclic codes of the given length, alphabet and twist; we therefore put the uniform measure on assignments, equivalently on the codes of this family itself (not on random linear codes). The mean and variance below are the exact ensemble averages of $\dim_{\mathbb{F}_q} \mathrm{Hull}_k$ over this family. Since $\mu_P/P \to 1/3$ as $P \to \infty$ (and $\mu_1 = 1/4$), the mean hull dimension is a positive proportion of the ambient dimension Δ — roughly $\Delta/3$ for large P, $\Delta/4$ at $P = 1$, apart from the length-one-orbit correction — so the typical code of the family has a hull growing linearly with n, in contrast to random linear codes, whose hull dimension stays bounded in probability [20]. The variance formulas quantify the fluctuations around this behaviour and, through the entanglement cost $c = n - K - H$ of the conditional quantum remark in Section X, describe its typical size in this ensemble. Comparisons with average-dimension formulas in the literature should be made only under matching hypotheses; the results below are stated strictly for the labeled repeated-root ensemble defined here.

Choose a labeled assignment uniformly at random; equivalently, all exponents A_i (over all components, orbits, positions) are independent and uniform on $\{0, \dots, P\}$. For one orbit write $X_i = \min(P - A_i, A_{i-1})$ (indices mod L) and $S_L = \sum_i X_i$; the orbit contributes $w_O S_{L_O}$ to $\dim_{\mathbb{F}_q} \mathrm{Hull}_k$ (Theorem V.2). Set

$$\mu_P = \frac{P(2P+1)}{6(P+1)}, \qquad \nu_P = \frac{\lfloor P^2/4 \rfloor}{P+1}.$$

Lemma VIII.1 (edge moments)**.** *For independent U, V uniform on $\{0, \dots, P\}$, $\mathbb{E}[\min(U,V)] = \mu_P$ and $\mathbb{E}[\min(U,V)^2] = \frac{P(P^2+P+1)}{6(P+1)}$. For $0 \le c \le P$, $F(c) := \mathbb{E}[\min(c,U)] = \frac{c(2P-c+1)}{2(P+1)}$.*

Proof. For a nonnegative integer variable, $\mathbb{E}[X] = \sum_{t \ge 1} \Pr(X \ge t)$ and $\mathbb{E}[X^2] = \sum_{t \ge 1} (2t-1) \Pr(X \ge t)$. Here $\Pr(\min(U,V) \ge t) = ((P - t + 1)/(P+1))^2$ for $1 \le t \le P$; summing via $\sum_{r=1}^{P} r^2 = P(P+1)(2P+1)/6$ and $\sum r^3 = (P(P+1)/2)^2$ gives both formulas. The last claim is $F(c) = \sum_{t=1}^{c} (P - t + 1)/(P+1)$. □

Theorem VIII.2 (means)**.** $\mathbb{E}[\dim_{\mathbb{F}_q} C] = \dfrac{n}{2} \sum_s m_s$, *and*

$$\mathbb{E}[\dim_{\mathbb{F}_q} \mathrm{Hull}_k(C)] \;=\; \nu_P \sum_{O:\, L_O = 1} w_O \;+\; \mu_P \sum_{O:\, L_O \ge 2} w_O L_O.$$

Proof. $\mathbb{E}[P - A_f] = P/2$ and $\sum_{f \in \mathcal{F}_s} \deg f = n_0$ give the first formula via Theorem V.2. For the hull, linearity over orbits reduces to $\mathbb{E}[S_L]$. If $L \ge 2$, each X_i is the minimum of two independent uniforms (for $L = 2$, X_0, X_1 are the two ordered pairings of the same two distinct variables), so $\mathbb{E}[S_L] = L\mu_P$ by Lemma VIII.1. If $L = 1$, $\mathbb{E}[S_1] = \frac{1}{P+1} \sum_{a=0}^{P} \min(P - a, a)$; the summands are symmetric under $a \mapsto P - a$, giving $P^2/4$ for $P = 2m$ and $(P^2 - 1)/4$ for $P = 2m + 1$, i.e. $\lfloor P^2/4 \rfloor$; hence $\mathbb{E}[S_1] = \nu_P$. Note $\nu_1 = 0$: at $P = 1$ an $L_O = 1$ orbit contributes nothing, and $\mathbb{E}[\mathrm{Hull}] = \frac{1}{4} \sum_{L_O \ge 2} w_O L_O$. □

Remark (relation to [5]). For a single component ($N = 1$, $K_s = \mathbb{F}_q$), the hull mean above agrees with [5, Theorem 3.3]: writing Δ_t for the sum of the degrees of the distinct factors in reciprocal cycles of length a_t (so

$\sum_t a_t\Delta_t = n_0$), their Formula simplifies, via $P - \frac{P(4P+5)}{6(P+1)} = \mu_P$ and $P - \frac{3P+1}{4} + \frac{\delta_P}{4(P+1)} = \nu_P$, to $\nu_P\Delta_1 + \mu_P\sum_{t\geq 2} a_t\Delta_t$, which is the display above. No variance or covariance formula is given there; the second-order results below (Theorem VIII.4 and Corollary VIII.5) have no counterpart in [5].

For the variance, orbit contributions are independent (disjoint exponent variables), so with $V_L(P) := \mathrm{Var}(S_L)$, $\mathrm{Var}(\dim_{\mathbb{F}_q}\mathrm{Hull}_k) = \sum_{s,O} w_O^2 V_{L_O}(P)$; note the squared weights, in contrast to the mean.

Lemma VIII.3 (variance pieces)**.** (i) *$L = 1$: $V_1(P) = \dfrac{P(P+2)(P^2+2P+4)}{48(P+1)^2}$ if P is even, and $V_1(P) = \dfrac{(P-1)(P+3)}{48}$ if P is odd.*

(ii) *For $L \geq 3$, two adjacent edges satisfy $\mathrm{Cov}(X_i, X_{i+1}) = \kappa_P < 0$, where $\kappa_P = -\frac{P(P+2)(7P^2+14P+9)}{360(P+1)^2}$, while non-adjacent edge pairs are independent.*

(iii) *For $L = 2$: $\mathbb{E}[X_0X_1] = \dfrac{P^2(P-1)}{12(P+1)}$ and $\mathrm{Cov}(X_0, X_1) = -\dfrac{P^2(P+2)^2}{36(P+1)^2}$.*

Proof. (i) $\mathbb{E}[\min(P-A, A)^2] = \frac{1}{P+1}\sum_a \min(P-a, a)^2$; for $P = 2m$ the squared values are $0^2, \ldots, (m-1)^2, m^2, (m-1)^2, \ldots, 0^2$, summing to $m(2m^2+1)/3$, and for $P = 2m+1$ each of $0^2, \ldots, m^2$ occurs twice, summing to $m(m+1)(2m+1)/3$; subtracting ν_P^2 (Theorem VIII.2) and simplifying gives the claim. (ii) X_i, X_{i+1} share exactly the variable A_i. Conditioning on $A_i = b$ makes them conditionally independent with means $F(P-b)$ and $F(b)$ (Lemma VIII.1), so $\mathbb{E}[X_iX_{i+1}] = \frac{1}{P+1}\sum_{b=0}^{P} F(P-b)F(b) = \frac{P(P-1)(11P^2+15P+6)}{120(P+1)^2}$ by the power sums $\sum b^j$, $j \leq 4$; subtracting μ_P^2 gives κ_P, negative for $P \geq 1$. Non-adjacent edges depend on disjoint pairs of variables, hence are independent. (iii) $X_0 \geq s$ and $X_1 \geq t$ hold simultaneously iff $s + t \leq P$, in which case each of the two variables has $P - s - t + 1$ admissible values; thus $\mathbb{E}[X_0X_1] = \sum_{s,t\geq 1}\Pr(X_0 \geq s, X_1 \geq t) = \frac{1}{(P+1)^2}\sum_{r=1}^{P-1}(P-r)r^2 = \frac{P^2(P-1)}{12(P+1)}$, and subtracting μ_P^2 gives the covariance. □

Theorem VIII.4 (variance of the hull dimension)**.** $\mathrm{Var}\left(\dim_{\mathbb{F}_q}\mathrm{Hull}_k(C)\right) = \displaystyle\sum_s \sum_{O\in\mathcal{O}_s} w_O^2\, V_{L_O}(P)$*, where*

$$
\begin{aligned}
&V_1(P) \text{ is as in Lemma VIII.3(i)},\\
&V_2(P) = \frac{P(P+2)(P^2+2P+3)}{18(P+1)^2},\\
&V_L(P) = \frac{L\,P(P+2)(P^2+2P+2)}{60(P+1)^2} \qquad (L \geq 3).
\end{aligned}
$$

Proof. Independence across orbits gives the weighted sum. For one orbit: $L = 1$ is Lemma VIII.3(i). For $L = 2$, $\mathrm{Var}(S_2) = 2\,\mathrm{Var}(X) + 2\,\mathrm{Cov}(X_0, X_1)$; with $\mathrm{Var}(X) = \frac{P(P+2)(2P^2+4P+3)}{36(P+1)^2}$ (from Lemma VIII.1) and (iii), the bracket factors as $P(P+2)(P^2+2P+3)$, giving $V_2(P)$. For $L \geq 3$, $\mathrm{Var}(S_L) = L\,\mathrm{Var}(X) + 2L\kappa_P$ (exactly the L adjacent pairs contribute, by (ii)), and $\mathrm{Var}(X) + 2\kappa_P = \frac{P(P+2)(P^2+2P+2)}{60(P+1)^2}$. At $P = 1$: $V_1(1) = 0$, $V_2(1) = 1/4$, $V_L(1) = L/16$. The $L = 2$ formula is not the $L \geq 3$ formula evaluated at $L = 2$ (at $P = 2$: $44/81$ versus $8/27$), because the two edges of a 2-orbit share both variables. □

Corollary VIII.5 (second-order structure of the two dimensions)**.** *Under the uniform model of this section,*

$$
\mathrm{Var}\left(\dim_{\mathbb{F}_q} C\right) = \frac{P(P+2)}{12}\sum_s \sum_{f\in\mathcal{F}_s} w_f^2, \qquad \mathrm{Cov}\left(\dim_{\mathbb{F}_q} C,\, \dim_{\mathbb{F}_q}\mathrm{Hull}_k(C)\right) = 0.
$$

In particular, the two dimensions are uncorrelated (they need not be independent).

Proof. By Theorem V.2, $\dim_{\mathbb{F}_q} C = \sum_{s,f} w_f(P - A_f)$ with independent A_f uniform on $\{0, \dots, P\}$, and $\mathrm{Var}(P - A_f) = P(P+2)/12$, giving the variance formula. For the covariance, orbit blocks are independent, so it suffices to consider one orbit with $K_O = \sum_i (P - A_i)$ and $S_O = \sum_j X_j$. In $\mathrm{Cov}(K_O, S_O)$, a term $\mathrm{Cov}(P - A_i, X_j)$ can be nonzero only for $i \in \{j, j-1\}$. With $\gamma_P := \mathrm{Cov}(B, \min(B, C)) = P(P+2)/24$ for independent B, C uniform on $\{0, \dots, P\}$ (a direct power-sum computation), the terms $i = j$ and $i = j - 1$ contribute $+\gamma_P$ and $-\gamma_P$ respectively when $L \geq 2$; for $L = 1$, symmetry under $A \mapsto P - A$ gives covariance 0 directly. Hence each orbit contributes 0, and summing with the weights w_O^2 gives $\mathrm{Cov} = 0$. □

IX. Examples and computational verification

The seven examples are arranged so that each one makes a different part of the theory visible. Example IX.1 is small enough to check by hand and shows how a single exponent tuple produces its two coordinates. Example IX.2 isolates the simple-root specialization. Examples IX.3 and IX.4 leave the orbit model and compare the exponent formulas with ranks of Gram matrices of explicitly constructed codes, first for mixed orbit lengths and then for a long orbit. Example IX.5 tests the product assembly over two components, and Example IX.6 leaves the cyclic case $\lambda = 1$ used in Examples IX.3–IX.5 and works with a compatible twist of order four. Example IX.7 solves one row of the EAQECC table below in full. Throughout, K denotes the code-dimension coordinate and H the hull-dimension coordinate of a labeled assignment.

Example IX.1 (one orbit, $P = 2$, $L = 2$, $w = 1$, by hand)**.** This example shows the bookkeeping of Theorem V.2 in the smallest repeated-root case. The two positions form one orbit, and for $\alpha = (\alpha_0, \alpha_1)$ the two coordinates are $K = \sum_i (2 - \alpha_i)$ and $H = \sum_i \min(2 - \alpha_i, \alpha_{i-1})$. For instance, for $\alpha = (0, 1)$ one gets $K = (2-0) + (2-1) = 3$ and $H = \min(2, 1) + \min(1, 0) = 1$; the first edge contributes because the predecessor exponent is positive, while the second edge contributes nothing because its predecessor exponent is 0. Evaluating all nine tuples gives

$$(0,0) \mapsto (4,0), \quad (0,1) \mapsto (3,1), \quad (0,2) \mapsto (2,2), \quad (1,0) \mapsto (3,1), \quad (1,1) \mapsto (2,2),$$
$$(1,2) \mapsto (1,1), \quad (2,0) \mapsto (2,2), \quad (2,1) \mapsto (1,1), \quad (2,2) \mapsto (0,0).$$

Grouping equal pairs yields the joint distribution $\{(4,0) : 1, (3,1) : 2, (2,2) : 3, (1,1) : 2, (0,0) : 1\}$. Two features are already visible. First, the multiplicities are symmetric under $K \leftrightarrow 4 - K$ at fixed H, as Corollary VI.4 requires for this orbit (here $\Delta = PwL = 4$). Second, the moments can be read off directly: $\mathbb{E}[S_2] = (0 \cdot 1 + 1 \cdot 2 + 2 \cdot 3 + 1 \cdot 2 + 0 \cdot 1)/9 = 10/9 = 2\mu_2$, and $\mathrm{Var}(S_2) = 44/81 = V_2(2)$, the $L = 2$ case of Theorem VIII.4. The separate $L = 2$ formula is essential here: the two edges of a 2-orbit share both variables, and formally inserting $L = 2$ into the $L \geq 3$ formula would give the incorrect value $8/27$. Finally, exactly two assignments, $\alpha \equiv 0$ and $\alpha \equiv P$, have hull contribution 0, as Corollary VII.3 requires per orbit.

Example IX.2 (simple-root orbit, $P = 1$, $L = 3$, $w = 1$)**.** At $P = 1$ the exponents are binary and the example checks the simple-root specialization of Proposition VI.5. The code coordinate is the number of zeros, $K = \sum_i (1 - \alpha_i)$, while the hull coordinate counts the cyclic transitions $1 \to 0$. The two constant words behave exceptionally:

000 gives $(K, H) = (3, 0)$ and 111 gives $(0, 0)$. Every non-constant binary word of length 3, viewed cyclically, has exactly one block of ones and one block of zeros, hence exactly one $1 \to 0$ transition and $H = 1$. There are three words of weight 1 (so $K = 2$) and three of weight 2 (so $K = 1$). Therefore the joint distribution is $\{(0,0) : 1,\ (1,1) : 3,\ (2,1) : 3,\ (3,0) : 1\}$, from which $\mathbb{E} = 3/4 = 3\mu_1$ and $\mathrm{Var} = 3/16 = V_3(1)$ follow immediately. The table also displays the symmetry of Corollary VI.4 in its smallest form: the hull multiplicity at $K = 0$ equals that at $K = 3$, and the multiplicity at $K = 1$ equals that at $K = 2$.

Example IX.3 (a repeated-root field case)**.** Take $K = \mathbb{F}_4$, $k = 1$, $\lambda = 1$, $P = 2$, $n_0 = 5$, $n = 10$. This case is chosen because it is genuinely repeated-root ($P = 2$) while still small, and because it contains both orbit types. The factors of $x^5 - 1$ over $\mathbb{F}_4$ have degrees $1, 2, 2$; the degree-one factor is fixed by τ, and the two degree-two factors form one orbit of length 2. Thus there are three exponent positions and $3^3 = 27$ assignments. (Dimensions in the display are over K itself; the orbit weights are therefore 1 for the length-one orbit and 2 for the length-two orbit.)

The point of the example is that the distribution below was computed in two independent ways: from the exponent formulas of Theorems V.2 and VI.3, and directly from codes, by forming, for each assignment a, the generator g_a, a basis of shifts, and the rank of the code-first Gram matrix $\big(\sum_j B_{t,j}\sigma(B_{r,j})\big)$. The two computations agree on the joint $(\dim_K C, \dim_K \mathrm{Hull}_1(C))$ distribution

$$\{(0,0) : 1, (1,1) : 1, (2,0) : 1, (2,2) : 2, (3,3) : 2, (4,2) : 2, (4,4) : 3, (5,5) : 3, (6,2) : 2,$$
$$(6,4) : 3, (7,3) : 2, (8,0) : 1, (8,2) : 2, (9,1) : 1, (10,0) : 1\},$$

whose entries sum to 27, as they must. The moment formulas can be understood orbit by orbit: the length-one orbit is a self-loop and contributes $\nu_2 = 1/3$, whereas the length-two orbit has weight 2 and two edges, hence contributes $2 \cdot 2\mu_2 = 20/9$. This gives $\mathbb{E}[\mathrm{Hull}] = 23/9 = \nu_2 + 2 \cdot 2\mu_2 = \frac{1}{3} + \frac{20}{9}$ — the separate ν_P term is not a cosmetic distinction here, since using μ_2 for the length-one orbit as well would give the wrong mean. The variance is $\mathrm{Var}(\mathrm{Hull}) = 194/81$ from Theorem VIII.4. The counting corollaries are equally visible: there are two orbits, so the LCD count is $4 = 2^2$ (Corollary VII.3), and the self-orthogonal and dual-containing counts agree at 12, in line with Corollary VII.5 and its refinement in Corollary VII.6.

Example IX.4 (orbit of length 6)**.** Take $K = \mathbb{F}_8$, $k = 1$, $\lambda = 1$, $P = 2$, $n_0 = 7$, $n = 14$. The seven linear factors of $x^7 - 1$ form τ-orbits of lengths 1 and 6, so there are $3^7 = 2187$ assignments. Whereas Example IX.3 tests mixed orbit lengths, this example is included for the length-six orbit. The Gram-rank computation was compared with the exponent formula assignment by assignment, not only through the aggregated distribution; it gives $\mathbb{E}[\mathrm{Hull}] = 11/3$ and $\mathrm{Var}(\mathrm{Hull}) = 10/9 = V_1(2) + V_6(2) = \frac{2}{9} + \frac{8}{9}$, with LCD count 4 and self-orthogonal = dual-containing count 258. The mean decomposition is again instructive: the length-one orbit contributes $\nu_2 = 1/3$, and the length-six orbit, of weight 1 with six edges, contributes $6\mu_2 = 10/3$.

For the record, the full joint distribution of $(\dim_{\mathbb{F}_q} C, \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C))$ is $(0,0)$: 1, $(1,1)$: 7, $(2,0)$: 1, $(2,2)$: 27, $(3,1)$: 6, $(3,2)$: 12, $(3,3)$: 59, $(4,2)$: 33, $(4,3)$: 48, $(4,4)$: 80, $(5,2)$: 24, $(5,3)$: 86, $(5,4)$: 96, $(5,5)$: 60, $(6,2)$: 24, $(6,3)$: 84, $(6,4)$: 144, $(6,5)$: 84, $(6,6)$: 21, $(7,2)$: 24, $(7,3)$: 84, $(7,4)$: 156, $(7,5)$: 102, $(7,6)$: 24, $(7,7)$: 3, $(8,2)$: 24, $(8,3)$: 84, $(8,4)$: 144, $(8,5)$: 84, $(8,6)$: 21, $(9,2)$: 24, $(9,3)$: 86, $(9,4)$: 96, $(9,5)$: 60, $(10,2)$: 33, $(10,3)$: 48, $(10,4)$: 80, $(11,1)$: 6, $(11,2)$: 12, $(11,3)$: 59, $(12,0)$: 1, $(12,2)$: 27, $(13,1)$: 7, $(14,0)$: 1.

The exponent formula is thus checked against the hull dimensions of actual codes on an orbit of length 6. One point must be stated correctly: a dimension check of this kind cannot certify the predecessor orientation, because the hull dimension does not depend on the orientation convention at all — for all a, b, $\min(P-a,b)-\min(P-b,a) = b-a$, which telescopes to 0 around any orbit, so the predecessor and successor readings give the same hull dimension for every assignment (cf. the remark after Theorem VI.3). The orientation is instead visible in the hull generator: here, the exponent of f_i in $h^{\#}$ equals the predecessor value $P - \alpha_{i-1}$ for every one of the 2187 assignments, while the successor value $P - \alpha_{i+1}$ agrees only for 27 of them (checked in `code/validate_audit_fixes.py`). The predecessor orientation is fixed by the code-first pairing (Theorem IV.3).

Example IX.5 (two components)**.** The final example tests the assembly step (Proposition VI.2 and Theorem VI.3), rather than a single orbit. Let $A = \mathbb{F}_2 \times \mathbb{F}_4$ ($q = 2$, $k = 0$, $\lambda = (1, 1)$, $P = 2$, $n_0 = 7$). The orbit data (L_O, w_O) are $(1, 1), (2, 3)$ on the first component and $(1, 2), (2, 6)$ on the second: four orbits and $1+2+1+2 = 6$ exponent positions in total, hence $3^6 = 729$ labeled assignments. Because assignments on distinct orbits are independent, the global joint distribution is the convolution (equivalently, the coefficientwise product) of the four per-orbit distributions of Theorem VI.1; the same distribution is obtained by computing the two components separately and adding their dimensions. This agreement is the concrete content of the product formula in Theorem VI.3.

The summary numbers illustrate how the global formulas are assembled. There are four orbits, so the LCD count is $2^4 = 16$ (Corollary VII.3). The self-orthogonal and dual-containing counts are 144 each, as Corollary VII.5 predicts. The mean splits according to orbit length: the two length-one orbits contribute $(1+2)\nu_2 = 1$, while the two length-two orbits contribute $(3{\cdot}2+6{\cdot}2)\mu_2 = 10$, giving $\mathbb{E}[\mathrm{Hull}] = 11$ (Theorem VIII.2). Likewise the variance is the weight-squared sum $\mathrm{Var}(\mathrm{Hull}) = 230/9$ of Theorem VIII.4, namely $1^2V_1(2) + 3^2V_2(2) + 2^2V_1(2) + 6^2V_2(2) = 230/9$. Thus Example IX.5 verifies, in one place, the independence of orbit blocks, the $L = 1$ versus $L \geq 2$ mean constants, and the the w_O^2 weighting in the variance.

Example IX.6 (a twist of order four)**.** All field-level examples so far have $\lambda = 1$. Take instead $K = \mathbb{F}_9 = \mathbb{F}_3(\iota)$ with $\iota^2 = -1$, $k = 1$, $\lambda = \iota$ of order 4 (compatible, since $\lambda^{1+3} = \lambda^4 = 1$), $P = 3$, $n_0 = 4$, $n = 12$. Here $\mu = -\iota$, and $x^4 - \mu = x^4 + \iota$ factors over $\mathbb{F}_9$ as two irreducible quadratics forming a single τ-orbit of length 2; hence there are $(3+1)^2 = 16$ assignments. The joint distribution of $(\dim_{\mathbb{F}_q} C, \dim_{\mathbb{F}_q} \mathrm{Hull}_k(C))$ is $\{(0,0) : 1, (2,2) : 2, (4,4) : 3, (6,6) : 4, (8,4) : 3, (10,2) : 2, (12,0) : 1\}$; hence $\mathbb{E}[\mathrm{Hull}] = 7/2$ and $\mathrm{Var}(\mathrm{Hull}) = 15/4$, the LCD count is $2 = 2^1$ (one orbit), and the self-orthogonal and dual-containing counts are 10 each. Every value was recomputed by Gram-rank brute force over the explicitly constructed codes (`code/validate_audit_fixes.py`), so the constacyclic theory is exercised here beyond the cyclic case $\lambda = 1$ and the negacyclic case $\lambda = -1$.

Example IX.7 (a worked EAQECC row)**.** We solve row 4 of Table II in full. Take $K = \mathbb{F}_4 = \mathbb{F}_2(\omega)$ with $\omega^2 + \omega + 1 = 0$, $k = 1$, $\lambda = 1$, $P = 2$, $n_0 = 3$, $n = 6$, and the assignment $a = (0, 1, 2)$ on the ordered factors $(x + 1, x + \omega, x + \omega^2)$ of $x^3 - 1$, so $g_a = (x + \omega)(x + \omega^2)^2$. Every root $\beta \in \{1, \omega, \omega^2\}$ satisfies $\beta^3 = 1$, hence $\beta^{-2} = \beta$; thus each linear factor is fixed by τ, giving three orbits of length 1 and weight 1. Theorem V.2 gives $K = (2-0) + (2-1) + (2-2) = 3$ and, each orbit being a self-loop, $H = \min(2,0) + \min(1,1) + \min(0,2) =$

$0+1+0=1$. The minimum distance is $d=3$ by the exact computation described in the computational-verification paragraph below, so $C(a)$ is a $[6,3,3]_4$ code. The conditional quantum remark of Section X therefore gives the EAQECC $[[6,K-H,d;\,n-K-H]]_4=[[6,2,3;2]]_4$, with EA Singleton defect $\Delta=2(n-K-d+1)=2$: the near-optimal (†) code of row 4.

a) A table of EAQECC-optimal and near-optimal codes.: We record ten further finite-field codes, all with $\lambda=1$ (hence $\mu=1$) and a single component ($N=m_1=1$), so all dimensions below are over $\mathbb{F}_q$ itself. Throughout, ω denotes a root of x^2+x+1 in $\mathbb{F}_4$, θ a root of x^3+x+1 in $\mathbb{F}_8$, and ι an element of $\mathbb{F}_9$ with $\iota^2=-1$. The exponent tuple a refers to the following ordered factor lists of $x^{n_0}-1$: over $\mathbb{F}_2$ with $n_0=3$: $(x+1,\ x^2+x+1)$; over $\mathbb{F}_3$ with $n_0=2$: $(x+1,\ x+2)$; over $\mathbb{F}_4$ with $n_0=5$: $(x+1,\ x^2+\omega x+1,\ x^2+\omega^2 x+1)$; over $\mathbb{F}_4$ with $n_0=3$: $(x+1,\ x+\omega,\ x+\omega^2)$; over $\mathbb{F}_5$ with $n_0=2$: $(x+1,\ x+4)$; over $\mathbb{F}_7$ with $n_0=2$: $(x+1,\ x+6)$; over $\mathbb{F}_8$ with $n_0=7$: $(x+\beta)_\beta$ with β running through $1,\theta,1+\theta,\theta^2,1+\theta^2,\theta+\theta^2,1+\theta+\theta^2$; over $\mathbb{F}_8$ with $n_0=3$: $(x+1,\ x^2+x+1)$; over $\mathbb{F}_9$ with $n_0=8$: $(x+\beta)_\beta$ with β running through $1,2,\iota,1+\iota,2+\iota,2\iota,1+2\iota,2+2\iota$; and over $\mathbb{F}_9$ with $n_0=2$: $(x+1,\ x+2)$. For each row, d is the minimum distance of $C(a)$, determined independently of the enumerator (exact computation; see the computational-verification paragraph below), and $H=\dim \mathrm{Hull}_k(C(a))$ is computed from Theorem V.2. The EAQECC column applies the conditional quantum remark of Section X, giving $[[n,K-H,d;\,n-K-H]]_q$. The last column is the EA Singleton defect $\Delta=\big(n-(K-H)+(n-K-H)\big)-2(d-1)=2(n-K-d+1)$. Following the convention of [7], we write * when $\Delta=0$, i.e. the EAQECC meets the EA Singleton bound (we call such a code EA-optimal), and † when $\Delta=2$ (near-optimal). Optimality here is meant only with respect to this EA Singleton bound; no comparison with best-known classical code tables is made. Rows with $P=1$ are simple-root specializations; the remaining rows are genuinely repeated-root.

TABLE II

TEN NEW FINITE-FIELD EXPONENT ASSIGNMENTS. THE EAQECC COLUMN GIVES THE OPTIMAL CODE $[[n,K-H,d;\,n-K-H]]_q$ OBTAINED FROM THE CONDITIONAL QUANTUM REMARK, MARKED * IF IT MEETS THE EA SINGLETON BOUND ($\Delta=0$) AND † IF $\Delta=2$.

No.	Setting	a	$[n,K,d]_q$	H	EAQECC	Δ
1	$\mathbb{F}_2$ (k=0, P=2, n_0=3)	$(2,0)$	$[6,4,2]_2$	0	$[[6,4,2;2]]_2^\dagger$	2
2	$\mathbb{F}_3$ (k=0, P=3, n_0=2)	$(1,3)$	$[6,2,4]_3$	1	$[[6,1,4;3]]_3^\dagger$	2
3	$\mathbb{F}_4$ (k=1, P=1, n_0=5)	$(0,0,1)$	$[5,3,3]_4$	2	$[[5,1,3;0]]_4^*$	0
4	$\mathbb{F}_4$ (k=1, P=2, n_0=3)	$(0,1,2)$	$[6,3,3]_4$	1	$[[6,2,3;2]]_4^\dagger$	2
5	$\mathbb{F}_5$ (k=0, P=5, n_0=2)	$(1,3)$	$[10,6,4]_5$	3	$[[10,3,4;1]]_5^\dagger$	2
6	$\mathbb{F}_7$ (k=0, P=7, n_0=2)	$(1,3)$	$[14,10,4]_7$	4	$[[14,6,4;0]]_7^\dagger$	2
7	$\mathbb{F}_8$ (k=1, P=1, n_0=7)	$(0,0,0,0,1,1,1)$	$[7,4,4]_8$	1	$[[7,3,4;2]]_8^*$	0
8	$\mathbb{F}_8$ (k=1, P=2, n_0=3)	$(2,0)$	$[6,4,2]_8$	0	$[[6,4,2;2]]_8^\dagger$	2
9	$\mathbb{F}_9$ (k=1, P=1, n_0=8)	$(0,1,0,0,0,1,1,1)$	$[8,4,5]_9$	2	$[[8,2,5;2]]_9^*$	0
10	$\mathbb{F}_9$ (k=1, P=3, n_0=2)	$(1,2)$	$[6,3,3]_9$	2	$[[6,1,3;1]]_9^\dagger$	2

b) Computational verification.: All formulas above were checked by exact (integer and rational) computation; the scripts and their saved outputs accompany the paper in the directory `code/`. (i) ***Orbit level*** (`code/validate_p1_14.py`): direct enumeration of exponent tuples against the transfer traces of Theorem VI.1, on the grid of 52 pairs (P,L)

with $P \in \{1,2,3,4,5,7,8,9\}$, $1 \le L \le 8$ and $(P+1)^L \le 10^5$; the moment closed forms were also re-derived symbolically from their defining sums. (ii) *Field level*: Gram-rank brute force over explicitly constructed codes for $\mathbb{F}_4$ with $n_0 = 3$ (27 assignments) and $n_0 = 5$ (27), $\mathbb{F}_9$ with $k = 1$, $\lambda = 1$ (256) and $\lambda = -1$ (256), $\mathbb{F}_3$ with $\lambda = -1$ (16), and $\mathbb{F}_8$ with $n_0 = 7$ (2187) — 2769 field-level assignments in total — together with the two-component assembly of Example IX.5 (729 assignments), checked by convolution of the component distributions. (iii) The twist of order four in Example IX.6 (16 assignments) and further field-level cases with $P \in \{4,8,9\}$ ($\mathbb{F}_2$ with $P = 4$ (25) and $P = 8$ (81); $\mathbb{F}_3$ with $P = 9$ (100)) are checked in `code/validate_audit_fixes.py`, which also verifies the hull-generator orientation statement of Example IX.4. (iv) The ten codes of Table II were checked separately by the accompanying script `code/validate_code_table.py`: for every row the hull dimension was computed both from Theorem V.2 and from the code-first Gram matrix, and the minimum distance exactly by parity-check column dependency, confirmed by an explicit minimum-weight codeword and, except in the $\mathbb{F}_7$ row, by direct enumeration of all codewords. These computations are supporting evidence for the tested cases; the proofs of Sections III–VIII are independent of them.

X. Limitations and scope

Scope. All enumerations count labeled exponent assignments over a fixed labeled decomposition $A \cong \prod_s K_s$; they are not counts of equivalence or isometry classes of codes, and no such interpretation should be attached to $\mathcal{E}(u,z)$. Only compatible twists are treated: for incompatible λ_s, Theorem IV.3 persists in the two-modulus form $x^n - \lambda_s^{-p^{d_s-k}}$, but the hull formula and everything following it are not asserted. The alphabet is reduced (square-free) by the standing hypothesis $A \cong \prod_s K_s$; non-reduced alphabets, such as chain rings, are outside the scope, as are distances, which the enumerator does not address. The means and variances of Section VIII refer to the uniform distribution on labeled assignments; orbit contributions are independent in this model, edges within an orbit are not (Lemma VIII.3), and the $L = 1$, $L = 2$, $L \ge 3$ regimes are genuinely distinct. Finally, the predecessor index α_{i-1} in Theorems IV.3–VI.3 is tied to the code-first pairing of Section II: reversing the convention replaces ρ by σ and α_{i-1} by α_{i+1}. By the telescoping identity $\min(P-a,b) - \min(P-b,a) = b - a$, every hull dimension — and hence every statistic of Sections VII and VIII — is unchanged assignment by assignment; the convention is visible only in the identification of the hull generator (Theorem V.2), not in any dimension.

Open problems. Natural directions include the incompatible-twist regime, where the dual exists in a two-modulus form but the hull intersection must be reconsidered; counting up to monomial equivalence rather than by labeled assignments; the extension from constacyclic (index-one) to multi-twisted codes; and distance distributions, which would require machinery beyond the enumerator.

Remark (conditional quantum remark). In the finite-field case ($N = 1$, $m_1 = 1$, so $q = p^e$ and $0 \le k < e$), let C be an $[n, K]$ code over $\mathbb{F}_q$ and let $H = \dim \mathrm{Hull}_k(C)$. Let Π be a parity-check matrix of C (rows a basis of $C^\perp$) and put $c = \mathrm{rank}\big(\Pi\, \rho(\Pi)^{\mathsf{T}}\big)$, with ρ applied entrywise. The kernel of the pairing map $D(C) \to (C^\perp)^*$, $y \mapsto \big(z \mapsto \sum_i z_i y_i\big)$, is exactly $C \cap D(C)$; hence $c = (n-K) - H$. Therefore, when the entanglement-assisted construction is used in its Galois form [7] (the framework of [3]), C yields an $[[n, K-H, d_{\mathrm{EA}}; n-K-H]]$ entanglement-assisted quantum code, where d_{EA} is the distance furnished by that construction and must be determined independently. The

enumerator of Theorem VI.3 determines the joint distribution of (K, H) only; it yields no distance information, and no quantum parameter is claimed from it alone. The product case $N \geq 2$ requires, in addition, control of distances under the product decomposition. The ten codes of Table II are explicit instances whose distances were computed independently, so their displayed parameters are unconditional; no asymptotic family is claimed, since for repeated-root families the distance — which the enumerator does not control — is the limiting quantity.

XI. Conclusion

For repeated-root constacyclic codes over a finite product of finite fields with compatible twists, the joint distribution of code dimension and Galois hull dimension over all labeled exponent assignments is given by a product of transfer-matrix traces indexed by the orbits of the reciprocal involution. The enumerator yields the total count, the code-dimension distribution, the LCD count $2^{\#\mathrm{orbits}}$, the self-orthogonal and dual-containing counts, and exact mean and variance formulas in which length-one orbits enter through the separate constant ν_P. At $P = 1$ the theory reduces to the binary simple-root enumerator. Natural directions for further work include the incompatible-twist regime, where the dual already exists in a two-modulus form but the hull intersection must be reconsidered; counting up to monomial equivalence rather than by labeled assignments; and distance questions, which the enumerator alone does not address.

References

[1] S. E. Anderson, E. Camps-Moreno, H. H. López, G. L. Matthews, D. Ruano, and I. Soprunov, Relative hulls and quantum codes, *IEEE Transactions on Information Theory* 70(5) (2024), 3190–3201. doi:10.1109/TIT.2024.3373550.

[2] S. Bouyuklieva, I. Bouyukliev, and F. Özbudak, Sequence of numbers of linear codes with increasing hull dimensions, *Designs, Codes and Cryptography* 94 (2026), Article 39. doi:10.1007/s10623-025-01776-9.

[3] T. Brun, I. Devetak, and M.-H. Hsieh, Correcting quantum errors with entanglement, *Science* 314 (2006), 436–439. doi:10.1126/science.1131563.

[4] B. Chen, H. Q. Dinh, and H. Liu, Repeated-root constacyclic codes of length $2\ell^m p^n$, *Finite Fields and Their Applications* 33 (2015), 137–159. doi:10.1016/j.ffa.2014.11.006.

[5] I. Debnath and O. Prakash, Average dimensions of Galois hulls of constacyclic codes, *Advances in Mathematics of Communications* 19(6) (2025), 1569–1604. doi:10.3934/amc.2025010.

[6] I. Debnath and O. Prakash, Hulls of separable double cyclic codes over $\mathbb{Z}_{p^2}$, *Computational and Applied Mathematics* 44 (2025), Article 255. doi:10.1007/s40314-025-03206-x.

[7] I. Debnath, H. Islam, E. Martínez-Moro, and O. Prakash, Galois hulls of constacyclic codes over affine algebra rings, *Discrete Mathematics* 349(2) (2026), Article 114750. doi:10.1016/j.disc.2025.114750; arXiv:2412.08512.

[8] I. Debnath, H. Islam, S. Yadav, and O. Prakash, Study of small Galois hull dimensions of constacyclic codes, *Advances in Mathematics of Communications* 22 (2026), 1–18. doi:10.3934/amc.2025054.

[9] I. Debnath, O. Prakash, and H. Islam, Galois hulls of constacyclic codes over finite fields, *Cryptography and Communications* 15(1) (2023), 111–127. doi:10.1007/s12095-022-00591-6.

[10] I. Debnath, O. Prakash, and H. Islam, Correction to: Galois hulls of constacyclic codes over finite fields, *Cryptography and Communications* 15 (2023), 129–130. doi:10.1007/s12095-022-00602-6.

[11] S. T. Dougherty and S. Şahinkaya, On cyclic and negacyclic codes with one-dimensional hulls and their applications, *Advances in Mathematics of Communications* 18(5) (2024), 1364–1378. doi:10.3934/amc.2022096.

[12] R. F. Taki Eldin and P. Solé, Generator polynomial matrices of the Galois hulls of multi-twisted codes, *Finite Fields and Their Applications* 109 (2026), Article 102712. doi:10.1016/j.ffa.2025.102712; arXiv:2305.15237.

[13] Y. Fan and L. Zhang, Galois self-dual constacyclic codes, *Designs, Codes and Cryptography* 84 (2017), 473–492. doi:10.1007/s10623-016-0282-8.

[14] W. C. Huffman and V. Pless, *Fundamentals of Error-Correcting Codes*, Cambridge University Press, 2003.

[15] H. Liu and X. Pan, Galois hulls of linear codes over finite fields, *Designs, Codes and Cryptography* 88 (2020), 241–255. doi:10.1007/s10623-019-00681-2.

[16] S. Li and M. Shi, Characterization and classification of binary linear codes with various hull dimensions from an improved mass formula, *IEEE Transactions on Information Theory* 70(5) (2024), 3357–3372. doi:10.1109/TIT.2023.3298379.

[17] S. Li, M. Shi, and S. Ling, A mass formula for linear codes with prescribed hull dimension and related classification, *IEEE Transactions on Information Theory* 71(1) (2025), 273–286. doi:10.1109/TIT.2024.3425058.

[18] Y. Liu, X. Yu, and G. Wang, Several constructions of linear codes with various Galois hull dimensions, *Computational and Applied Mathematics* 44 (2025), Article 299. doi:10.1007/s40314-025-03261-4.

[19] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977.

[20] N. Sendrier, On the dimension of the hull, *SIAM Journal on Discrete Mathematics* 10(2) (1997), 282–293. doi:10.1137/S0895480195294027.

[21] A. Sharma, Repeated-root constacyclic codes of length $\ell^t p^s$ and their dual codes, *Cryptography and Communications* 7 (2015), 229–255. doi:10.1007/s12095-014-0106-5.

[22] S. Talbi, A. Batoul, A. Fotue Tabue, and E. Martínez-Moro, Galois hulls of cyclic serial codes over a finite chain ring, arXiv:2102.06995 (2021).

[23] T. Wu, S. Zhu, L. Liu, and L. Li, Repeated-root constacyclic codes of length $6\ell^m p^n$, *Advances in Mathematics of Communications* 17(5) (2023), 1154–1180. doi:10.3934/amc.2021044.

[24] E. Zhang, B. Kong, and X. Zheng, Quantum codes from Galois hulls of constacyclic codes over a finite non-chain ring, *Entropy* 28 (2026), 407. doi:10.3390/e28040407.

[25] Q. Zhang, W. Wang, S. Luo, and Y. Li, Repeated-root constacyclic codes of length $k^s \ell^m p^n$ over finite fields, *Finite Fields and Their Applications* 101 (2025), Article 102542. doi:10.1016/j.ffa.2024.102542.